Course: Big Data

Lab 04

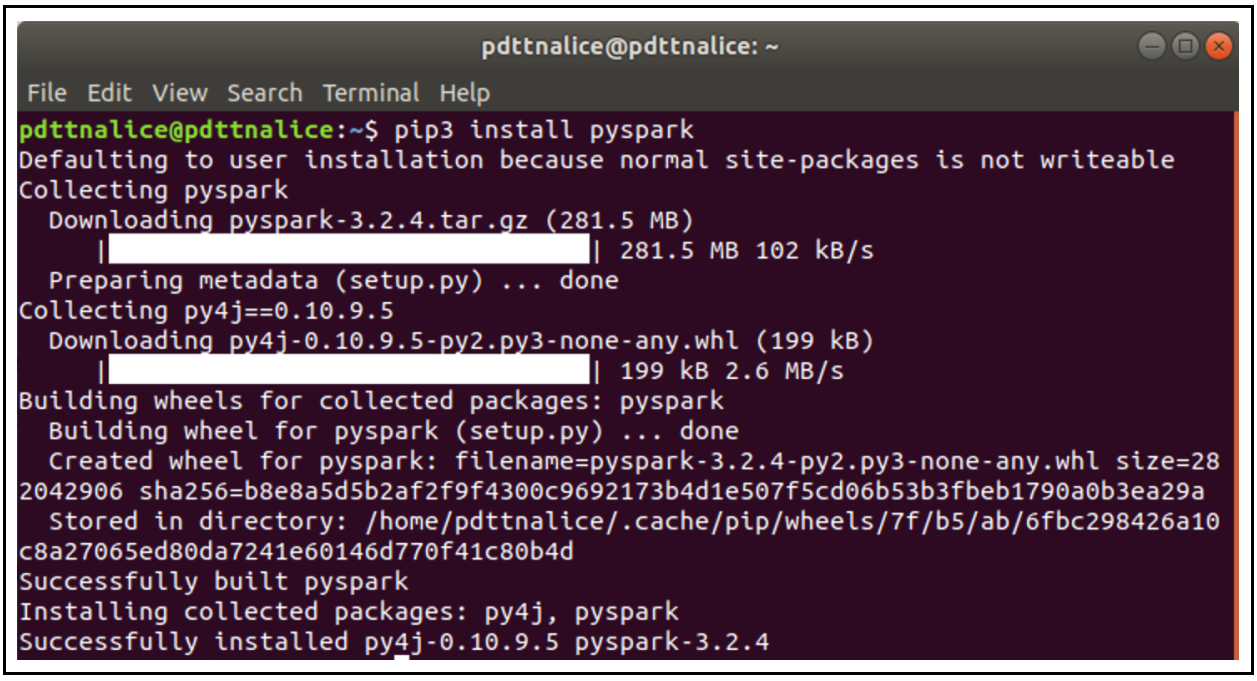
PySpark - RDD

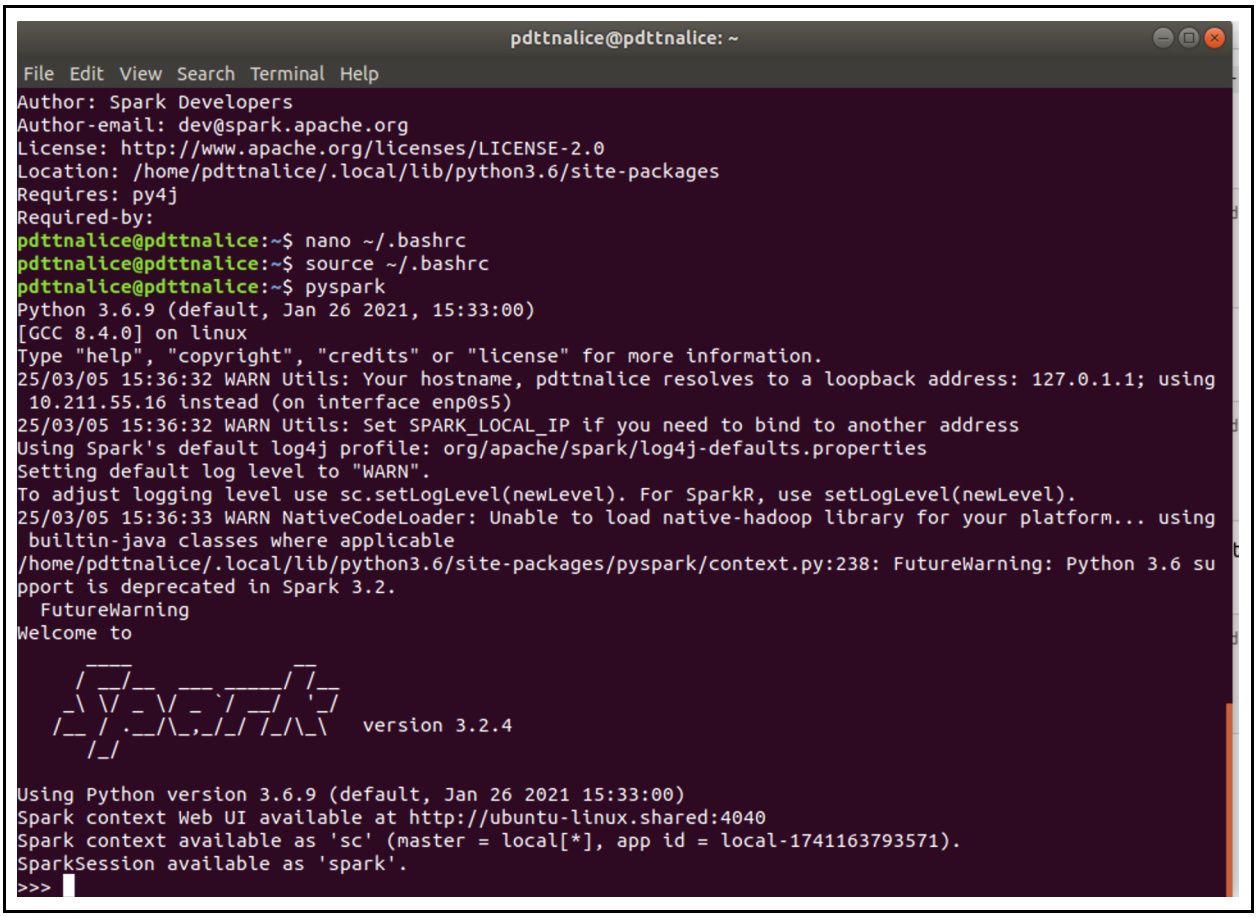
522H0148 - Phạm Đặng Thanh Trung

Question 1:

Based on [the tutorial of PySpark], students install PySpark in Ubuntu.

- Define the environment variable: JAVA_HOME
- Define the environment variable: SPARK_HOME
- Start the pyspark-shell and write an instruction to print down the PySpark version
- Take the screenshot and insert it into the table below.



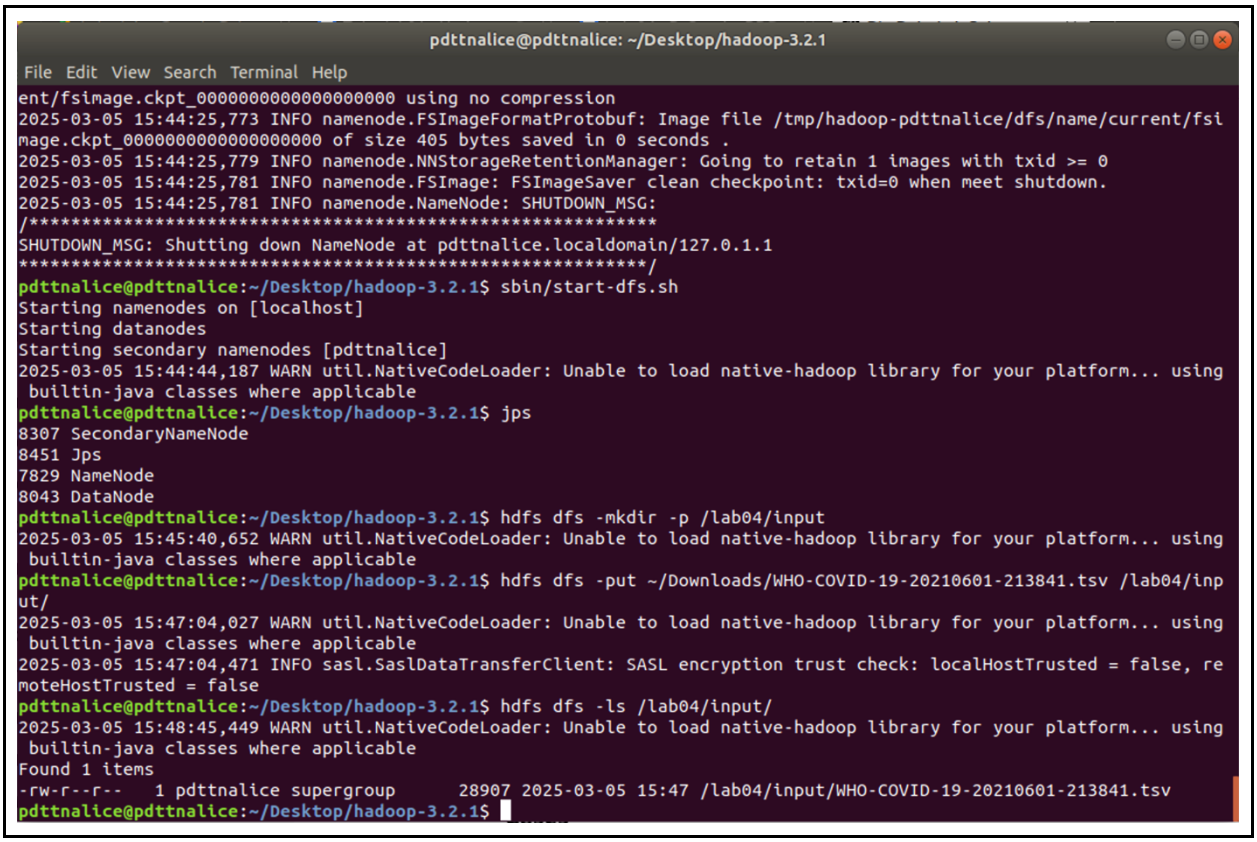


Question 2:

Given a tsv file WHO-COVID-19-20210601-213841.tsv which is corresponding to the WHO Coronavirus (COVID-19) Dashboard.

Students are required to create a folder, named **lab04**, in HDFS and then copy the tsv to **lab04/input/**

Take a screenshot to show the content of **lab04/input/** in HDFS



Question 3:

Write a PySpark program, located in **ASEANCaseCount.py**, to count the number of cumulative total cases among ASEAN countries (*South-East Asia Region in the given data table*) using RDDs.

- Insert your source code into the table below.

```
from pyspark import SparkContext, SparkConf
from pyspark.sql import SparkSession

#Initialize Spark Session
conf=SparkConf().setAppName("ASEAN Case Count")
sc=SparkContext(conf=conf)
spark=SparkSession(sc)

#Đọc file TSV từ HDFS vào RDD
rdd=sc.textFile("hdfs://localhost:9000/lab04/input/WHO-COVID-19-20210601-213841.tsv")

#Chia dữ liệu và lọc cho các quốc gia ASEAN
def parse_line(line):
    fields=line.split("\t")
    return fields[0].strip(), fields[1].strip(), fields[2].strip(), fields[3].strip()  #Điều chỉnh theo cấu trúc file của bạn
```

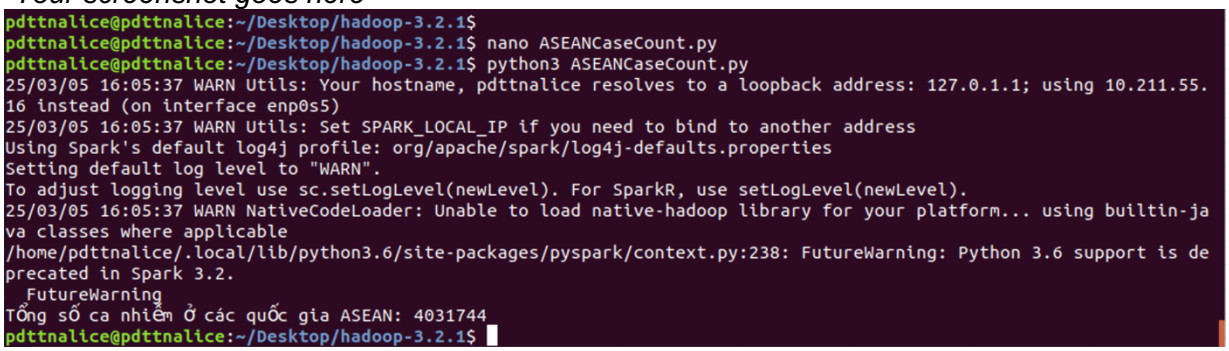
```
#Danh sách các quốc gia ASEAN
ASEAN_countries=["Brunei Darussalam", "Cambodia", "Indonesia", "Lao People's Democratic Republic",
                 "Malaysia", "Myanmar", "Philippines", "Singapore", "Thailand", "Viet Nam"]

#Lọc ra các quốc gia ASEAN
filtered_rdd=rdd.map(parse_line).filter(lambda x: x[0] in ASEAN_countries)

#Kiểm tra nếu RDD bị rỗng
if filtered_rdd.isEmpty():
    print("Không có dữ liệu cho các quốc gia ASEAN.")
else:
    #Chuyển đổi dữ liệu và tính tổng số ca nhiễm
    total_cases=filtered_rdd.map(lambda x: int(float(x[2].replace(",", "")))).reduce(lambda x, y: x + y)
    print(f"Tổng số ca nhiễm ở các quốc gia ASEAN: {total_cases}")
```

- Take a screenshot of the terminal to visualize the program result.

Your screenshot goes here



Submission Notice

- Export your answer file as pdf
- Rename the pdf following the format:

  lab04_<student number>_<full name>.pdf

  E.g. lab04_123456_NguyenThanhAn.pdf

  If you have not been assigned a student number yet, then use 123456 instead.
- Careless mistakes in filename, format, question order, etc. are not accepted (0 pts).